

Angular Routing – Technical Documentation

Subgroup One-Liners

- Router Setup – Configures route definitions that map URL paths to corresponding application components using the routing module.
- Navigation – Implements client-side route transitions programmatically or declaratively within a single-page architecture.
- Guards – Enforces conditional route activation through guard interfaces that validate access control logic before navigation.
- Authentication Flows – Manages user identity verification and maintains authenticated state to control protected route access.
- Error Handling – Handles undefined or unmatched routes using fallback configurations to ensure controlled navigation behavior.

Internal Modules

- Routes Configuration – Defines structured route objects mapping URL patterns to components.
- Default Route – Establishes a fallback entry path when no specific route is matched.
- Redirect Route – Automatically forwards navigation to a specified target path.
- Wildcard Route – Captures unmatched routes to prevent undefined navigation states.
- Router Initialization – Bootstraps the routing infrastructure within the application configuration.
- RouterLink Directive – Enables declarative template-based navigation.
- Programmatic Navigation – Triggers route transitions using router APIs.
- Relative Navigation – Resolves route paths based on the current route context.
- SPA Navigation Handling – Maintains stateful transitions without full browser reloads.
- CanActivate Guard – Validates route access before component activation.
- Access Control Logic – Evaluates authorization conditions during navigation.
- Conditional Redirection – Redirects users when guard validation fails.
- Route Protection Strategy – Integrates security checks directly into route definitions.
- Credential Validation – Verifies user authentication data against defined rules.
- Session Persistence Mechanism – Maintains authentication state across sessions.
- Post-Login Redirection – Navigates authenticated users to protected areas.
- Logout Handling – Clears authentication state to restrict protected access.
- Fallback Route Mechanism – Directs unmatched URLs to a controlled error view.
- 404 Error Component – Displays structured feedback for invalid navigation requests.

- Route Ordering Strategy – Ensures fallback routes execute after valid routes.
- Navigation Fail-Safe Handling – Prevents unstable routing states during invalid access.